Section: **3.2 – MECHANICALLY FASTENED JOINTS**

Part: **3210**

Title: **TYPES OF RIVETED JOINTS**

Revision: **1.5**

Date: **30/07/2024**

1 INTRODUCTION

This chapter classifies the different typical types of mechanically fastened joints between structural components and directs the Stress Engineer to the most appropriate method to calculate the loads in the fasteners.

Sub-chapter §2 details how to estimate the stiffness of each fastener, which is part of the input data in those calculations. Following sub-chapters present the different configurations of fastened joints that can be found in airframe, and refer to the available methodologies for obtaining the fasteners loads in each case. If the case of study cannot be assimilated within any of the types discussed, typically a Finite Elements Model will be required.

In addition to abovementioned material, guidance on how global internal loads (either available from global finite elements models (GFEM) or other means) can be derived into the specific loads sets necessary for calculating each type of bolting group can be found in chapter 3215 “INPUT FORCES FOR FASTENERS GROUPS ANALYSIS”.

2 STIFFNESS OF A FASTENER

2.1 STIFFNESS OF A FASTENER IN SHEAR LOAD

Commonly accepted Huth’s formula for calculating the fastener flexibility (*C* in figure below) shall be used to obtain the fastener stiffness subjected to shear loads.

$$C = \left(\frac{t_1 + t_2}{2d}\right)^a \frac{b}{n}\left(\frac{1}{t_1E_1} + \frac{1}{nt_2E_2} + \frac{1}{2t_1E_3} + \frac{1}{2nt_2E_3}\right)$$

Group I—bolted metallic joints	$a = 2/3;\ b = 3.0$
Group II—riveted metallic joints	$a = 2/5;\ b = 2.2$
Group III—bolted graphite/epoxy joints	$a = 2/3;\ b = 4.2$

Figure 1. Huth's formula for fastener flexibility (see Ref. 1, original, and translation Ref. 2)

Where t_1 and t_2 are the thicknesses of the plates, d the rivet (shank) diameter, E_1, E_2 are the Young moduli of plates (1,2) and E_3 the Young modulus of the bolt/rivet. Values of parameters a and b are stated in previous figure. Parameter n is equal to 1 in single shear joints and equal to 2 in double shear joints. Notice plate 1 is the central plate in double shear configurations.

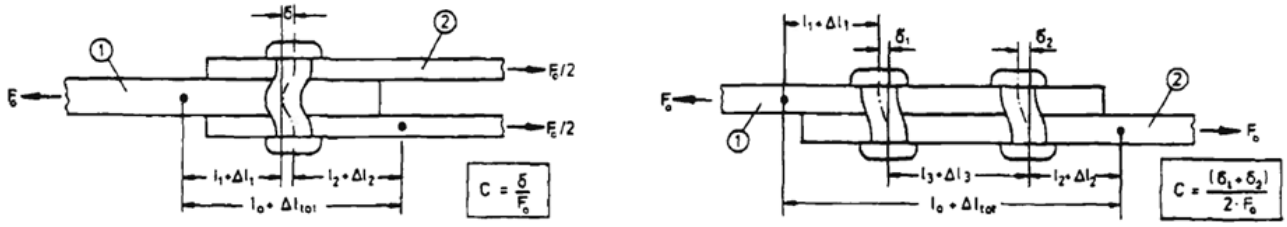

Figure 2. Double Shear and Single Shear configurations

In case of bolted hybrid joints, if one of the plates is metallic and the other made of a composite laminate, the following natural extension of the abovementioned formula shall apply:

$$C = \left(\frac{t_1 + t_2}{2d}\right)^a \frac{1}{n}\left(\frac{b_1}{t_1E_1} + \frac{b_2}{nt_2E_2} + \frac{b_1}{2t_1E_3} + \frac{b_2}{2nt_2E_3}\right)$$

… with exponent a equal to 2/3 , b_1 and b_2 shall be equal to 3.0 when the affected plate i is metallic , or 4.2 when the plate i is made of a (carbon/epoxy) composite material, and rest of parameters with the same meaning discussed previously.

The fastener stiffness is the inverse of the flexibility, thus:

$$K = 1 / C$$

Alternative expressions are proposed and used by other aerospace companies and institutions, as those described in Huth's paper, or those in ESDU98012. The Huth's formula presented in this chapter is preferred, however, in most cases.

Remarks:

- ✓ For some kind of analyses, with rivets in double shear configuration, it may of interest to model the attachment with one independent spring between each pair of consecutive plates.

  If the flexibility of the double-shear joint, using Huth's notation, is related to the total load (taken by the centre plate) and the total displacement, with the expression:

$$C = \delta / F \qquad \text{(see Figure 2)}$$

That for each of the 2 “springs” joining each of the external plates with the central plate, installed “in parallel” respect to that central plate, shall be:

$$C' = \delta' / F' = (\delta / 2) / (F / 2) = C$$

… then the same expression used for the total load and displacement can be used for the individual springs joining the external plates. Changing the notation of the plates from numerical indices (in Huth’s paper) to subindices *o* for referring the outer plate and *i* for the inner plates, the Huth’s formula for the flexibility in double shear becomes:

$$C_o = [\,(t_i + t_o) / (2 \cdot d)\,]^a \cdot (1/2) \cdot [\, b_i / (t_i \cdot E_i) + b_o / (2 \cdot t_o \cdot E_o) + b_i / (2 \cdot t_i \cdot E_3) + b_o / (4 \cdot t_o \cdot E_3)\,]$$

And the stiffness associated to the single springs in that double shear configuration:

$$K_o = 1 / C_o$$

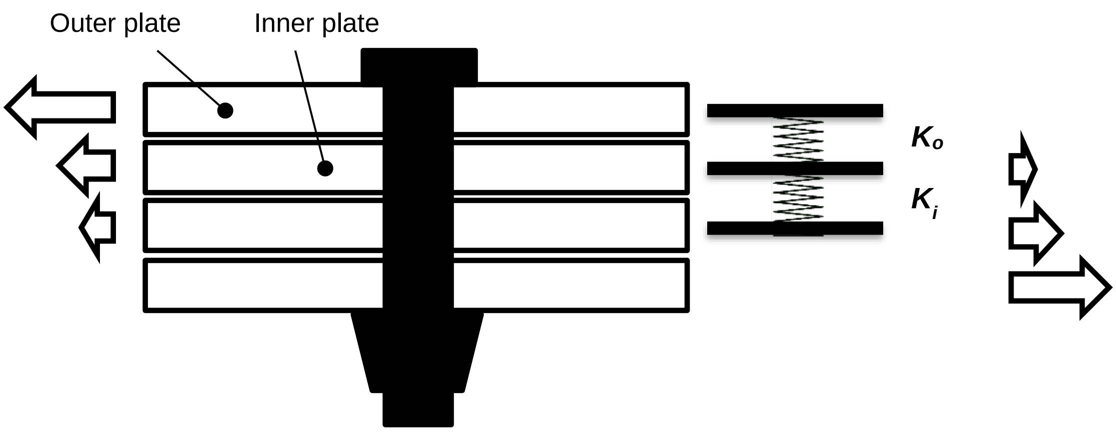

Figure 3. Sketch of multi-plate fastened joint

But… notice the expression is not symmetrical, and would lead to different values depending on the order selected for the plates if the intended spring stiffness searched is of an interface between two internal plates, in joints of 4 or more plates. Indeed, in that expression for estimating the flexibility the terms associated to the central plate are half those of the external plates (central plate contributes with more stiffness). The correction proposed for the formula to fix this issue is:

$$C_i = [(t_{i1} + t_{i2}) / (2 \cdot d)]^a \cdot (1/2) \cdot [\, b_{i1} / (t_{i1} \cdot E_{i1}) + \boldsymbol{b_{i2} / (t_{i2} \cdot E_{i2})} + b_{i1} / (2 \cdot t_{i1} \cdot E_3) + \boldsymbol{b_{i2} / (2 \cdot t_{i2} \cdot E_3)}\,]$$

$$K_i = 1 / C_i$$

To be confirmed, eg. with high fidelity analysis (DFEM) if this approach is not overestimating the stiffness of those joints.

✓ In some types of fastened attachments, rivets are installed intentionally with high clearance (floatability), with the purpose of reducing the loads induced by the deformation of the surrounding structure: fairings, covers on access holes,… In FEM non-linear analysis the expected behaviour of this type of joints can be modelled using non-linear elements (gaps) including, or connected to, spring elements. The selection of an appropriate stiffness for linear elastic analyses may not be simple. When the main driver of the loads taken by the joint is the deformation of the surrounding structure the part is attached to, using the yield/failure point as reference to derive an apparent linear stiffness would be reasonable.

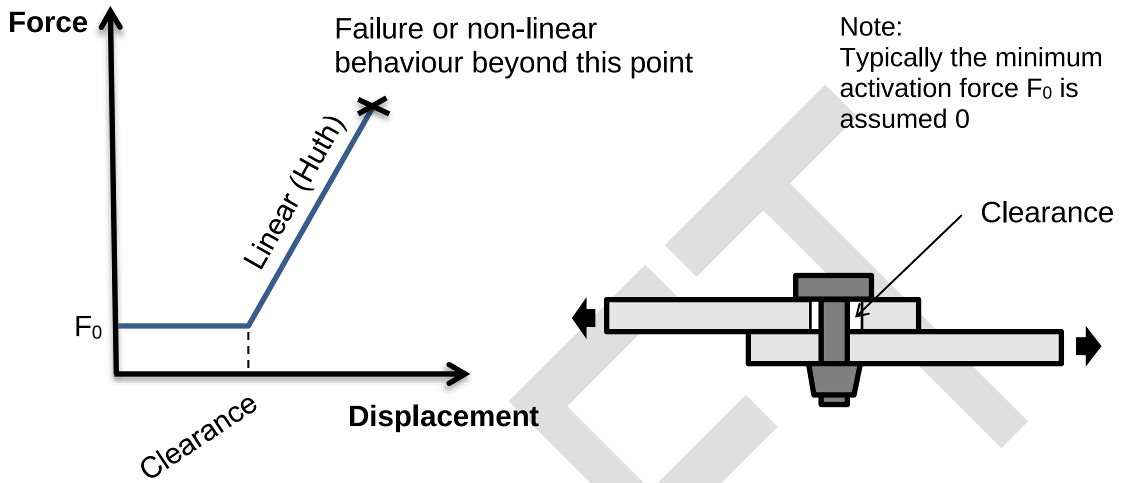

Figure 4. Fastened joint in-plane stiffness in installations with clearance

2.2 AXIAL STIFFNESS OF A FASTENER IN TENSION LOAD

Typically fasteners are modelled as spring elements in those Finite Element Models where plates are modelled with 2D elements. In detailed FEMs with solid elements modelling all parts of the structure, approximations of the fastener stiffness are not necessary. For those FEM's built using bi-dimensional elements and spring elements connecting the plates, the axial stiffness of the rivet in tension is easily estimated from its cross-sectional inertia and its grip (*):

Flexibility of rivet $C = \Delta L / F = \varepsilon \cdot L / F = L / EA = \Sigma(t) / EA$

Stiffness of rivet $K = 1/ C = EA / L = EA / \Sigma(t)$

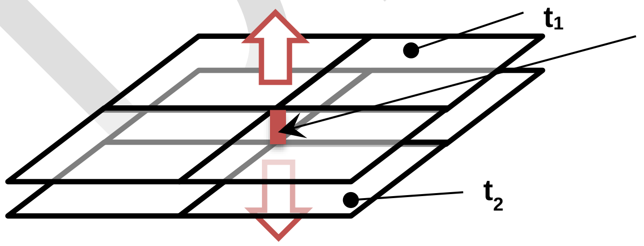

Figure 5. Sketch of FEM detail with spring element joining plates (2D modelled)

(*) Other company divisions recommend to use, as reference rod length to derive the rivet axial stiffness, the sum of the grip plus 1/3 of rivet height plus 1/3 of nut height : $L = \Sigma(t) + H_{head}/3 + H_{nut}/3$

In case more than 2 plates and more 1 spring element are involved in the joint, it shall not be difficult to derive the corresponding axial stiffness of each spring (normally the intermediate plates will contribute in just half its thickness to the total length of each of the two spring elements connected to them). In order to prevent/mitigate the clash between plates in this type of models, it is also recommended to complete the fastened connection between the 2D plate elements with rotational springs (of dummy-very-high stiffness), especially in linear analyses without contact elements between the plates.

Nevertheless, analytical methods for calculating the rivets loads in eccentric joints typically assume the fitting is a rigid body, not explicitly computing the flexibility of the plates attached due to bending deformation. This will lead to overestimate the stiffness at the fastener location with rivet in tension loads. This problem can be mitigated by adding the bending contribution within the base-plates to the flexibility of the fastener itself :

$$C = C_{fastener} + \Sigma(C_{plates}) = \Sigma(t) / EA + \Sigma[\, e^3 / (E \cdot w \cdot t^3) \,]$$

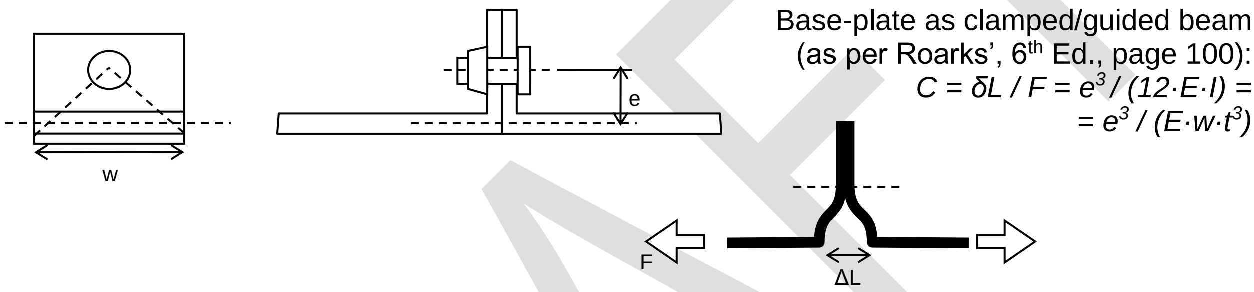

Figure 6. Typical arrangement of rivet in a tension fitting

For those methodologies (when plates bending is not explicitly computed), the applicable stiffness in axial direction (rivet-tension load) can then be set to :

$$K = 1 / C = 1 / \{\, \Sigma(t) / EA + \Sigma[\, e^3 / (E \cdot w \cdot t^3) \,] \,\}$$

Remarks:

- ✓ The contribution of the term corresponding to the bolt axial stiffness (Σ(t)/EA) is questionable in pre-tensioned joints (from no applied load up to the load of separation of plates)
- ✓ Rivets separated a big distance from any transversal wall (lever arm *e* is high) will not be effective in reacting tension loads

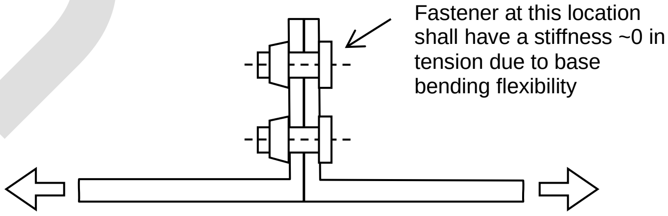

Figure 7. Fastener separated from wall not effective for reacting tension forces

2.3 BENDING AND ROTATIONAL STIFFNESS OF FASTENERS

Certain types of analyses, e.g. with Finite Element Models in which fasteners are idealized with spring elements, shall require design values for the stiffness provided by rivets against bending and axial rotation.

When the contact between the joined parts is not modelled, a minimum value for the bending stiffness is recommended. For simplicity, *the preferred method is to use a fixed magnitude of* 10^9 Nmm for this value (Ref. 6).

An arbitrarily small value of the rotational stiffness around the fastener axis of 100 Nmm is recommended. The relative rotation between the two plates attached, around the axis normal the plane, shall be naturally sustained by shear loads in the bolts group the rivet is part of.

Alternative formulations to estimate these values can be found in Ref. 6 and Ref. 7.

3 CONTINUOUS JOINTS

For calculating the rivets loads in a continuous joint (e.g. fastened joints between skin panels, or between a doubler and the stringers ends it is connecting), where the stress problem is solved in one dimension (no loads redistributions in other directions), the method explained in chapter 3211 “RIVETS LOADS IN CONTINUOUS JOINTS” shall be used.

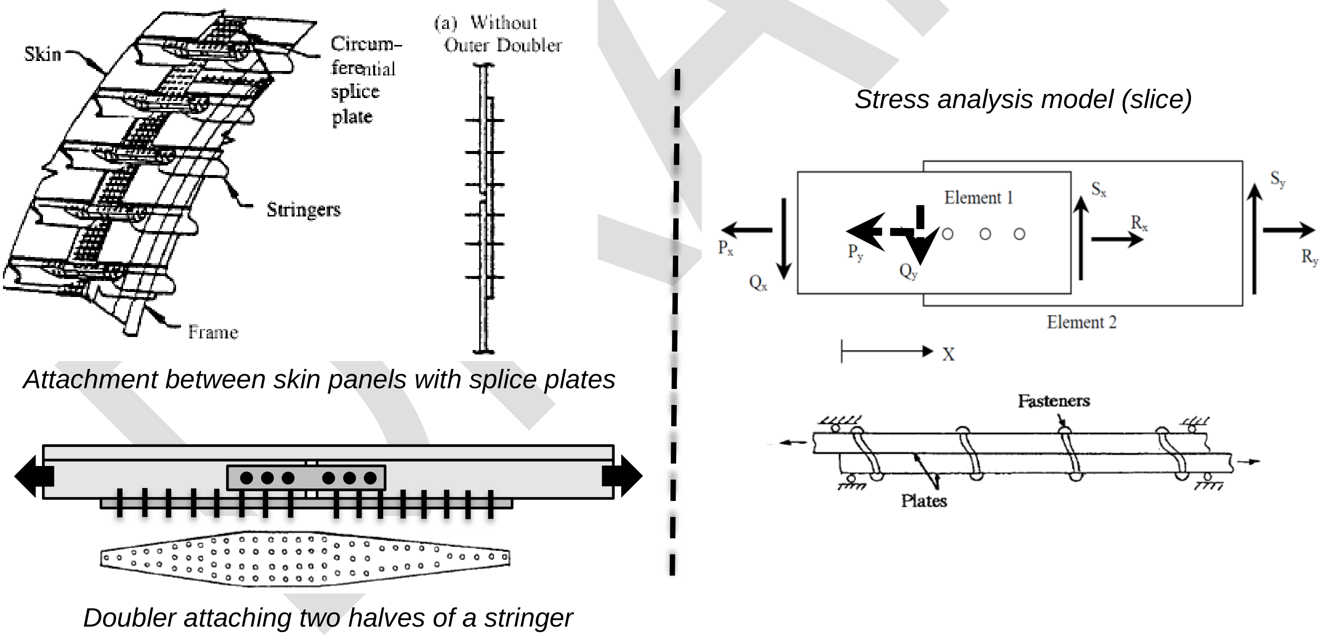

Figure 8. Typical examples of continuous joints and sketch of the stress analysis model

4 PLANE JOINTS WITH IN-PLANE LOADS

For calculating the rivets loads in a plane joint with in-plane loads and potential load redistributions within the plane (e.g. skin repair patches), typically a FEM analysis is necessary. Under some circumstances some simplifications can be done to deal with these problems using non-numerical methodologies. Review chapter 3212 “RIVETS LOADS IN PLANE JOINTS WITH IN-PLANE LOADS” for further details.

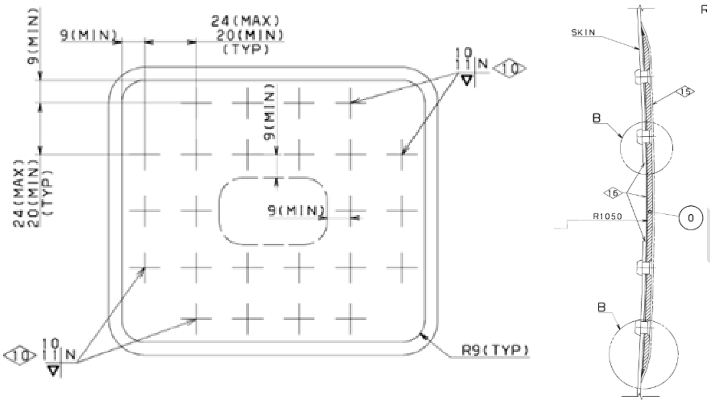

Figure 9. Extract of drawing of a typical bolted repair in a skin panel

5 PLANE JOINTS WITH ECCENTRIC LOADS

For calculating the loads in the bolts attaching a fitting (a “rigid-body”) to a support structure, if all fasteners are installed in the same plane, use method explained in chapter 3213 “RIVETS LOADS IN PLANE JOINTS WITH ECCENTRIC LOADS”.

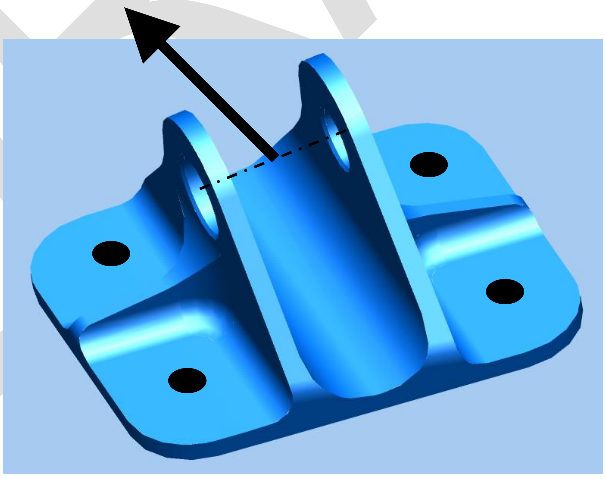

Figure 10. Fitting with plane bolted attachment

6 3D JOINTS WITH ECCENTRIC LOADS

For calculating the loads in the bolts attaching a fitting (a “rigid-body”) to a support structure, with fasteners installed in different planes, use method explained in chapter 3214 “RIVETS LOADS IN ECCENTRIC 3D JOINTS”.

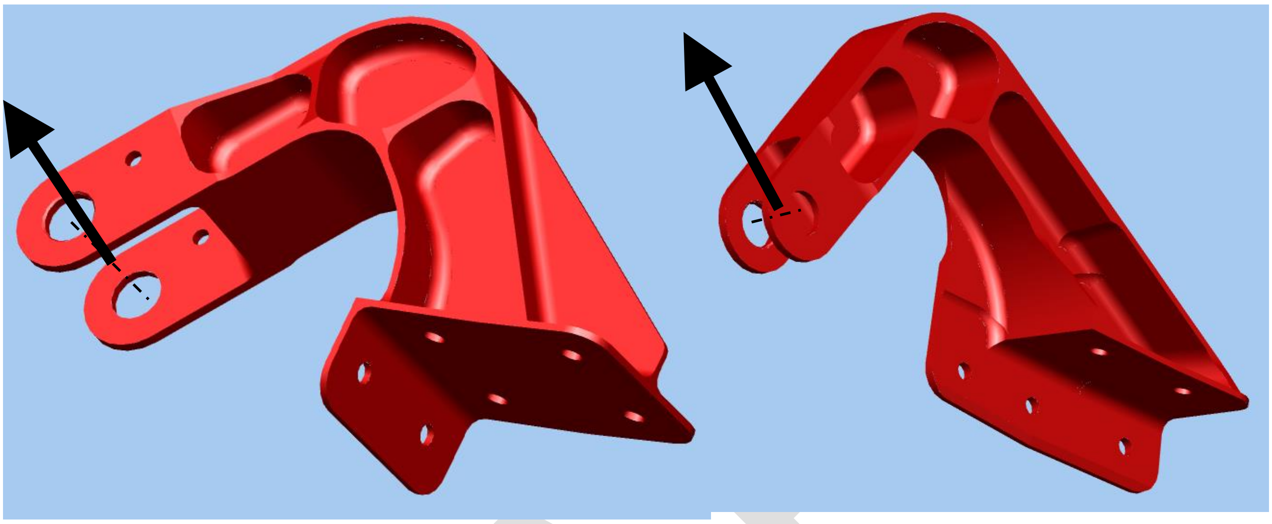

Figure 11. Fitting with non-plane bolted attachment

7 SCOPE AND LIMITATIONS

Calculation of rivets/bolts loads in a mechanically fastened joint between two (or more) structural components.

See referred chapters for details on specific limitations of each of the methods proposed.

It is remarked that the expressions proposed to estimate the stiffness in rivet axial direction (in tension loads) at a fastener location, for application of non-numerical (non-FEM) methodologies, is just an approximation which might lead to non-negligible errors when calculating the rivets loads. The Stress Engineer shall evaluate the consistency of the assumptions done with the real design, check the availability of similar parts where the methodology was validated (by test or simulation) and, in case of doubts (especially if strength check results in short margins), prepare a more precise analysis model using FEM simulation.

8 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
1	-	As subscript, refers to plate 1 (central plate in double shear joint)
2	-	As subscript, refers to plate 2
3	-	As subscript, refers to the rivet/bolt
δ, δL , ΔL	mm	Partial or total deformation/elongation
ε	-	Strain
a, b	-	Parameters of the Huth's formula
A	mm^2	Area of the cross-section
C	mm/N	Fastener flexibility
d	mm	Diameter of the fastener/hole
e	mm	Lever arm
E	MPa	Young modulus
F	N	Force
FEM	-	Finite Elements Model
H	mm	Height (of rivet head or nut in this context)
K	N/mm	Fastener stiffness
L	mm	Length (rivet/bolt grip in this context)
n	-	Parameter of the Huth's formula, equal to 2 for configuration sin double shear joints, and equal to 1 in single lap shear
N/A	-	Not Applicable
t	mm	Thickness of the plate
w	mm	Effective width

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[H. Huth]	1984	Zum Einfluß der Nietnachgiebigkeit mehrreihiger Nietverbindungen auf die Lastübertragungs und Lebensdauervorhersage
Ref. 2	[H. Huth]	1986	Influence of Fastener Flexibility on the Prediction of Load Transfer and Fatigue Life for Multiple-Row Joints
Ref. 3	ESDU98012	10/2001	Flexibility of, and load distribution in, multi-bolt lap joints subject to in-plane axial loads
Ref. 4	[Warren C. Young]	6^{th} Ed.	Roark's Formulas for Stress and Strain
Ref. 5	[M. C. Y. Niu]	1995	AIRFRAME STRUCTURAL DESIGN
Ref. 6	[D. Madier]	1^{st} Ed.	Practical Finite element analysis for mechanical engineers
Ref. 7	[Airbus] V53PR0906133	Rev. 2.1	A350 Detailed FEM Modelling guidelines for Linear and Non Linear Analysis

SOFTWARE

Program	***Version***	***Description***
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures (module *JOINTSTIFF*)

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
1.0 01/08/2018	G. Baños	First issue	All pages
1.1 04/03/2020	G. Moreda	Section number corrected and Approval Signatures table updated according to new template	All pages
1.2 29/04/2020	G. Baños	Coverage to double shear joints modelled as 2 independent springs. Removed distribution list of chapter and software table updated.	Pages 2, 8
1.3 29/04/2021	G. Baños	Applicability of Huth's formula extended to hybrid (metal/composite) joints	Page 2
1.4 03/10/2022	G. Baños	Bending and rotational stiffness of fasteners	Page 5
1.5 30/07/2024	G. Baños	Huth formula adapted to fastened joints of multiple plates (greater than 2)	Page 2, 3

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Methods & Optimization	G. Baños 30/07/2024	-
Structural Analysis	-	F. Glock DD/MM/2024

Table 3. *Approval signatures table for last revision of this chapter*